

Solution STA510 Exam spring 2026

Problem 1

Let X be a real-valued random variable with density function

$$f(x) = \frac{1 + \sin(x)}{\pi + 2}, \quad 0 \leq x \leq \pi,$$

and $f(x) = 0$ otherwise.

a)

Show that $f(x)$ is a proper probability density function.

For $x \in [0, \pi]$, we have $\sin(x) \geq 0$, hence $1 + \sin(x) \geq 1 > 0$, and $f(x) = 0$ otherwise. Further we have

$$\begin{aligned} \int_0^\pi \frac{1 + \sin(x)}{\pi + 2} dx &= \frac{1}{\pi + 2} \left(\int_0^\pi 1 dx + \int_0^\pi \sin(x) dx \right) \\ &= \frac{1}{\pi + 2} \left([x]_0^\pi + [-\cos(x)]_0^\pi \right) \\ &= \frac{1}{\pi + 2} \left((\pi - \cos(\pi)) - (-\cos(0)) \right) \\ &= \frac{1}{\pi + 2} (\pi + 2) = 1 \end{aligned}$$

Therefore $f(x)$ is a proper pdf.

b)

For the CDF, we have

$$\begin{aligned} F(x) = P(X \leq x) &= \int_0^x \frac{1 + \sin(y)}{\pi + 2} dy = \frac{1}{\pi + 2} \left([y]_0^x + [-\cos(y)]_0^x \right) \\ &= \frac{1}{\pi + 2} (x - \cos(x) + \cos(0)) = \frac{x - \cos(x) + 1}{\pi + 2} \end{aligned}$$

and

$$F(X) = U \Leftrightarrow \frac{x - \cos(x) + 1}{\pi + 2} = U \Leftrightarrow x - \cos(x) = (\pi + 2)U - 1$$

This mixes the variable x both linearly and inside a trigonometric function. Equations of this form have no algebraic or closed-form solution.

c)

We wish to generate i.i.d. samples from $f(x)$ using the Accept-Reject (AR) method with a Uniform proposal. i) Specify the proposal distribution, and find the envelope constant c for the AR method. ii) Compute the numerical value of the average acceptance probability. iii) Write down the AR algorithm (in pseudocode) to generate one draw from $f(x)$ using your proposal distribution and constant c .

i) As X is random variable supported on $[0, \pi]$, we need to propose values for $[0, \pi]$, i.e. we propose values from a Uniform $U(0, \pi)$ distribution with density

$$g(x) = \frac{1}{\pi}, \quad x \in [0, \pi]$$

.

Then, we need to find a constant c with

$$\frac{f(x)}{g(x)} = \frac{1 + \sin(x)}{\pi + 2} \pi \leq c.$$

From calculus, we know that $\sin(x)$ is maximized at $x = \frac{\pi}{2}$ on $[0, \pi]$, and $\sin(\pi/2) = 1$. Thus,

$$c = \max_x \frac{f(x)}{g(x)} = \frac{1 + \sin(\pi/2)}{\pi + 2} \pi = \frac{2\pi}{\pi + 2}.$$

ii) The AR algorithm:

1. Draw $Y \sim U(0, \pi)$
2. Draw $U \sim U(0, 1)$
3. Compute acceptance ratio

$$\alpha(Y) = \frac{f(Y)}{cg(Y)} = \frac{1 + \sin(Y)}{\pi + 2} \pi \frac{\pi + 2}{2\pi} = \frac{1 + \sin(Y)}{2}$$

4. If $U \leq \alpha(Y)$, accept and set $X = Y$. Otherwise, reject Y and go back to step 1.

Problem 2

a)

Consider the following R code:

```
T <- 1000
x <- numeric(T)
x[1] <- 0

for (t in 2:T) {
  x[t] <- x[t - 1] + sample(c(-1, 1), size = 1, prob = c(0.5, 0.5))
}
```

i) Identify what kind of stochastic process this R code simulates. Justify your answer. ii) Is this a Markov process? Justify your answer! iii) How would the process change if you modify $c(0.5, 0.5)$ to $c(0.4, 0.6)$?

i) The code simulates a one-dimensional symmetric random walk, a type of discrete-time stochastic process where each step is equally likely to move up $+1$ or down -1 .

ii) This process has the Markov property, because the next position depends only on the current position, not on previous history.

iii) The process becomes an asymmetric (biased) random walk, introducing a drift toward the more probable direction (in this case, upward drift if $+1$ has probability 0.6).

b)

Write down an expression for the stochastic process shown in the above R code. What is the state space of this process? Discuss whether this process has a stationary distribution. Justify your reasoning analytically by finding the expectation and variance of this process given $E(X_1) = 0$.

This process can be express as

$$X_t = X_{t-1} + Z_t \text{ where } Z_t \in \{-1, 1\}, \quad P(Z_t = 1) = P(Z_t = -1) = \frac{1}{2}.$$

This process does not have a stationary distribution. The random walk is not bounded; its state space is the entire set of integers, $X_t \in \mathbb{Z}$. Further, we can shown that the variance grows without bound. We have

$$E(Z_t) = (-1) \cdot P(Z_t = -1) + (1) \cdot P(Z_t = 1) = -\frac{1}{2} + \frac{1}{2} = 0.$$

and

$$E(X_t) = E(X_{t-1} + Z_t) = E(X_{t-1}) + E(Z_t) = E(X_{t-1}) = 0$$

because $E(X_1) = 0$. Further, we can write $X_t = Z_1 + Z_2 + \dots + Z_t$ where Z_i 's are independent. Thus

$$Var(X_t) = Var(Z_1 + Z_2 + \dots + Z_t) = Var(Z_1) + \dots + Var(Z_t) = t \cdot 1 = t$$

because

$$Var(Z_i) = \sum_{z_i} (z_i - \mu)^2 f(z_i) = ((-1) - 0)^2 \cdot P(Z_t = -1) + ((1) - 0)^2 \cdot P(Z_t = 1) = \frac{1}{2} + \frac{1}{2} = 1.$$

Hence, $Var(X_i) = t$, i.e. for increasing t the mean of the process remains 0, whereas the variance increases linearly with t , i.e. the variance grows without bounds, and thus the random walk “spreads out” as it progresses and has no stationary distribution.

Problem 3

We want to estimate the mathematical constant π based on the property that $P(X^2 + Y^2 \leq 1) = \frac{\pi}{4}$ for $X \sim U[0, 1]$ and $Y \sim U[0, 1]$. For this we have implemented two functions in the following R-code:

```
# Option 1:
estPi1 <- function(n=10000){
  y <- numeric(n)
  for(i in 1:n){
    u <- runif(2)
    y[i] <- (u[1]^2+u[2]^2 <= 1)
  }
  return(4*mean(y))
}

# Option 2:
```

```
estPi2 <- function(n=10000){
  u <- matrix(runif(2*n),nrow=n,ncol=2,byrow = T)
  y <- mean(rowSums(u^2) <= 1)
  return(4*y)
}
```

a)

Which algorithm has been implemented for estPi1()? Reason your answer, and describe the choices made for the algorithm. Write down an expression for the estimator $\hat{\pi}$ of π .

The implemented algorithm for estPi1() is a crude MC estimator. We have

$$\hat{\pi} = 4 \int_0^1 \int_0^1 \mathbb{I}_{x^2+y^2 \leq 1}(x, y) f(x, y) dy dx = 4 \int_0^1 \int_0^1 \mathbb{I}_{x^2+y^2 \leq 1}(x, y) dy dx$$

with $f(x, y)$ the density function of 2-dimensional uniform distribution on $[0, 1]^2$,

$$f(x, y) = \begin{cases} 1 & , 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 1 \\ 0 & , \text{otherwise} \end{cases}$$

and the indicator function

$$\mathbb{I}_{x^2+y^2 \leq 1}(x, y) = \begin{cases} 1 & , x^2 + y^2 \leq 1 \\ 0 & , \text{otherwise.} \end{cases}$$

Given $X_1, \dots, X_n \sim U[0, 1]$ and $Y_1, \dots, Y_n \sim U[0, 1]$ the crude Monte Carlo estimator is given by

$$\hat{\pi}_{CMC} = 4 \cdot \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{X_i^2 + Y_i^2 \leq 1}(X_i, Y_i).$$

This is what is implemented in the function estPi1().

b)

Do both of the two above R-functions estPi1() and estPi2(), return a correct estimate for π ? Justify your answer.

Both algorithm are correct implementation of $\hat{\pi}_{CMC}$ to estimate π . The difference is that estPi1() is an un-vectorized implementation, and estPi2() is a vectorized implementation. Thus, the 2nd implementation is faster.

c)

Explain how we can improve estPi1() to estimate π , and describe the necessary changes in the R-code.

We could improve the Monte Carlo estimator based on 5000 pairs of antithetic variables $V = \mathbb{I}_{X^2+Y^2 \leq 1}(X, Y)$ and $\tilde{V} = \mathbb{I}_{(1-X)^2+(1-Y)^2 \leq 1}(1-X, 1-Y)$ with $X \sim U[0, 1]$ and $Y \sim U[0, 1]$ by

$$\hat{\pi}_{AntMC} = 4 \cdot \frac{1}{n} \sum_{i=1}^{n/2} \left(\mathbb{I}_{X_i^2 + Y_i^2 \leq 1}(X_i, Y_i) + \mathbb{I}_{(1-X_i)^2 + (1-Y_i)^2 \leq 1}(1-X_i, 1-Y_i) \right) \quad \text{with } n = 10000.$$

There are different possible implementations. One implementation is to add `v <- 1-u` after `u <- runif(2)`, and to replace `y[i] <- ...` by `y[i] <- ((u[1]^2+u[2]^2 <= 1) + (v[1]^2+v[2]^2 <= 1))/2`, i.e. the code looks like this

```
estPi1 <- function(n=10000){
  y <- numeric(n)
  for(i in 1:n){
    u <- runif(2)
    v <- 1-u
    y[i] <- ((u[1]^2+u[2]^2 <= 1) + (v[1]^2+v[2]^2 <= 1))/2
  }
  return(4*mean(y))
}
```

Problem 4

Assume we are monitoring the air pollution in Stavanger day by day, and each day we collect multiple sensor readings $y_1, \dots, y_n$ where y_i describes the concentraion of the air pollution at time point $i = 1, \dots, n$, and n is the total number of readings for today.

Now, we want to estimate for the current day, the true (unobserved) mean pollution level μ with within-day variability σ^2 while incorporating prior knowledge (from past days) about what typical pollution levels and variances look like. It is assumed that

$$y_i \mid \mu, \sigma^2 \sim N(\mu, \sigma^2), \quad i = 1, \dots, n.$$

For the prior information about μ and σ^2 , we are assuming that the mean population level

$$\mu \mid \sigma^2 \sim N(m_0, \sigma^2/\kappa_0)$$

with known $m_0, \kappa_0 > 0$, and for the variance we are assuming

$$\log(\sigma^2) \sim N(0, 1)$$

independently of μ .

a)

Show that the joint posterior kernel $p(\mu, \sigma^2 \mid y_1, \dots, y_n)$ is given by

$$p(\mu, \sigma^2 \mid y_1, \dots, y_n) \propto (\sigma^2)^{-\frac{(n+3)}{2}} \exp \left(-\frac{1}{2\sigma^2} \left(S(\mu) + \kappa_0(\mu - m_0)^2 \right) - \frac{(\log(\sigma^2))^2}{2} \right)$$

with $S(\mu) = \sum_{i=1}^n (y_i - \mu)^2$.

Let $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$ and $S(\mu) = \sum_{i=1}^n (y_i - \mu)^2$, then the Likelihood is given by

$$\begin{aligned} L(\mu, \sigma^2 \mid y_1, \dots, y_n) &= \prod_{i=1}^n f(y_i \mid \mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y_i - \mu)^2} \\ &\propto \frac{1}{(\sigma^2)^{n/2}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2} = (\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2} S(\mu)} \end{aligned}$$

The prior for $\mu \mid \sigma^2$ is given by

$$p(\mu \mid \sigma^2) \propto (\sigma^2)^{-1/2} e^{-\frac{\kappa_0}{2\sigma^2}(\mu - m_0)^2}$$

The prior for σ^2 is given by $\log(\sigma^2) \sim N(0, 1) \Leftrightarrow \sigma^2 \sim \log N(0, 1)$. The density function for a log-normal distribution is given by

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2 x}} \exp\left(-\frac{(\log(x) - \mu)^2}{2\sigma^2}\right).$$

Thus for $\log N(0, 1)$ we have $f(x) = \frac{1}{\sqrt{2\pi x}} \exp\left(-\frac{(\log(x))^2}{2}\right)$ and therefore the prior density kernel for σ^2 is given by

$$p(\sigma^2) \propto (\sigma^2)^{-1} e^{-\frac{(\log(\sigma^2))^2}{2}}.$$

Hence

$$\begin{aligned} p(\mu, \sigma^2 \mid y_1, \dots, y_n) &\propto (\sigma^2)^{-n/2} (\sigma^2)^{-1/2} (\sigma^2)^{-1} e^{-\frac{1}{2\sigma^2}(S(\mu) + \kappa_0(\mu - m_0)^2) - \frac{(\log(\sigma^2))^2}{2}} \\ &= (\sigma^2)^{-(n+3)/2} e^{-\frac{1}{2\sigma^2}(S(\mu) + \kappa_0(\mu - m_0)^2) - \frac{(\log(\sigma^2))^2}{2}} \end{aligned}$$

b)

Derive the conditional log posterior kernel for $\log(p(\mu \mid \sigma^2, y_1, \dots, y_n))$, and show that the conditional posterior $\mu \mid \sigma^2, y_1, \dots, y_n \sim N(m_n, \sigma^2/\kappa_n)$. Find m_n and κ_n !

The conditional for $\mu \mid \sigma^2$ (condition on σ^2) uses only terms involving μ . Thus,

$$\begin{aligned} \log p(\mu \mid \sigma^2, y_1, \dots, y_n) &\propto \log\left((\sigma^2)^{-(n+3)/2} e^{-\frac{1}{2\sigma^2}(S(\mu) + \kappa_0(\mu - m_0)^2) - \frac{(\log(\sigma^2))^2}{2}}\right) \\ &\propto -\frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i - \mu)^2 + \kappa_0(\mu - m_0)^2\right) \\ &= -\frac{1}{2\sigma^2} \left(\sum_{i=1}^n (y_i^2 - 2y_i\mu + \mu^2) + \kappa_0(\mu^2 - 2m_0\mu + m_0^2)\right) \\ &= -\frac{1}{2\sigma^2} \left(\left(\sum_{i=1}^n y_i^2 + \kappa_0 m_0^2\right) - 2\left(\kappa_0 m_0 + \sum_{i=1}^n y_i\right)\mu + (n + \kappa_0)\mu^2\right) \end{aligned}$$

Given $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$, and the relation that if $\log(p(x)) = a + bx + cx^2$, then $x \sim N(-\frac{b}{2c}, -\frac{1}{2c})$ from the formulas sheet, we have

$$b = \frac{2(\kappa_0 m_0 + n\bar{y})}{2\sigma^2} = \frac{(\kappa_0 m_0 + n\bar{y})}{\sigma^2}, \quad c = -\frac{n + \kappa_0}{2\sigma^2}$$

and thus

$$-\frac{b}{2c} = \frac{(\kappa_0 m_0 + n\bar{y}) \cdot 2\sigma^2}{2 \cdot \sigma^2 \cdot (n + \kappa_0)} = \frac{n\bar{y} + \kappa_0 m_0}{n + \kappa_0} = m_n$$

and

$$-\frac{1}{2c} = \frac{2\sigma^2}{2 \cdot (n + \kappa_0)} = \frac{\sigma^2}{n + \kappa_0} = \frac{\sigma^2}{\kappa_n}$$

with $\kappa_n = n + \kappa_0$. And thus, $\mu \mid \sigma^2, y_1, \dots, y_n \sim N(m_n, \sigma^2/\kappa_n)$.

c)

Derive the conditional posterior kernel for $\sigma^2 \mid \mu, y_1, \dots, y_n$, and explain why it is not a standard distribution we can sample from directly.

The conditional for $\sigma^2 \mid \mu$ (condition on μ) uses only terms involving σ^2 . Let $K(\mu) = S(\mu) + \kappa_0(\mu - m_0)^2$, then

$$\begin{aligned} \log p(\sigma^2 \mid \mu, y_1, \dots, y_n) &\propto \log \left((\sigma^2)^{-(n+3)/2} e^{-\frac{1}{2\sigma^2} (S(\mu) + \kappa_0(\mu - m_0)^2) - \frac{(\log(\sigma^2))^2}{2}} \right) \\ &= -\frac{n+3}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} K(\mu) - \frac{1}{2} (\log(\sigma^2))^2 \end{aligned}$$

This is not a standard distribution with a simple expression for the log-density function. The log-density kernel looks close to a inverse-gamma log-density, but the presence of the term $\log(\sigma^2))^2$ breaks it. Thus, we need to resort to a Metropolis Hasting step.

d)

Let us consider a Metropolis Hasting step for the posterior kernel for $\sigma^2 \mid \mu, y_1, \dots, y_n$ with proposal

$$\log(\sigma^2) \sim N(\sigma_{t-1}^2, s^2).$$

Specify the proposal distribution $q(\cdot \mid \cdot)$, and derive the Metropolis–Hastings acceptance probability $\log(\alpha)$.

Here, we have two possibilities.

Option 1 (directly proposing σ^2):

If $\log(\sigma_*^2) \sim N(\log(\sigma_{t-1}^2), s^2)$, then $\sigma_*^2 = e^{\log(\sigma_*^2)} \sim \log N(\log(\sigma_{t-1}^2), \tau^2)$, i.e.

$$q(\sigma_*^2 \mid \sigma_{t-1}^2) = \log N(\log(\sigma_{t-1}^2), \tau^2)$$

This proposal takes only positive values and is not symmetric. Thus we need to include the lognormal proposal ratio in the MH acceptance. Given $\log p(\sigma^2) = \log p(\sigma^2 \mid \mu, y_1, \dots, y_n)$ as above in c), we have

$$\log(\alpha) = \min \left\{ 0, \left(q(\sigma_{t-1}^2 \mid \sigma_*^2) + \log p(\sigma_*^2) \right) - \left(q(\sigma_*^2 \mid \sigma_{t-1}^2) + \log p(\sigma_{t-1}^2) \right) \right\}$$

Option 2 (simulating $\eta = \log(\sigma^2)$):

If we do a variable shift by $\eta = \log(\sigma^2)$, we need to include the Jacobian factor into the transformed density. If $\eta = \log(\sigma^2)$, then $\sigma^2 = e^\eta$, and $d\sigma^2 = e^\eta d\eta$. Thus,

$$\pi(\eta \mid \mu, y_1, \dots, y_n) = p(e^\eta \mid \mu, y_1, \dots, y_n) e^\eta$$

$$\begin{aligned} \log \pi(\eta \mid \mu, y_1, \dots, y_n) &= \log p(e^\eta \mid \mu, y_1, \dots, y_n) + \eta \\ &\propto -\frac{n+3}{2} \eta - \frac{1}{2e^\eta} K(\mu) - \frac{1}{2} \eta^2 + \eta \\ &= -\frac{n+1}{2} \eta - \frac{1}{2e^\eta} K(\mu) - \frac{1}{2} \eta^2 \end{aligned}$$

Thus, after variable shift, the MH step simplifies to RWMH algorithm with symmetric proposal

$$q(\eta^* \mid \eta_{t-1}) = N(\eta_{t-1}, s^2)$$

and

$$\log(\alpha) = \min \left\{ 0, \log \pi(\eta^*) - \log \pi(\eta_{t-1}) \right\}$$

Finally, we have to transform back our sample by $\sigma_*^2 = e^{\eta^*}$.

e)

Describe briefly the Gibbs sampler algorithm to simulate values for $p(\mu, \sigma^2 \mid y_1, \dots, y_n)$.

Given some initial values (μ_0, σ_0^2) , the Gibbs sampler algorithm alternates between

- sampling from μ_t from its closed-form conditional posterior,

$$\mu_t \sim N(m_n, \sigma_{t-1}^2 / \kappa_n).$$

- and updating σ_t^2 given μ_t and $K(\mu_t)$ using the MH step as described in d).